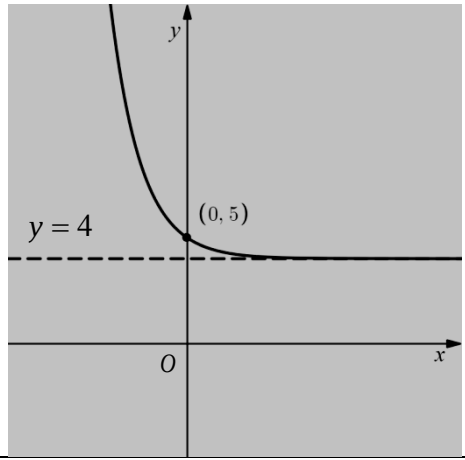
1	$f(x) = ax^3 + 3x^2 - 8x + 2$	
	Sets $f(2) = 3 \Rightarrow 8a + 12 - 16 + 2 = 3$	M1
	$\Rightarrow 8a = 5 \Rightarrow a = \frac{5}{8}$	M1A1
		(3)
		(3 marks)

2	$\left(\frac{3}{8}+4x\right)^{12}$	
	Term in x^7 is $^{12}C_7\left(\frac{3}{8}\right)^5\left(4x\right)^7$ or coefficient of x^7 is $^{12}C_7\left(\frac{3}{8}\right)^5 4^7$	M1 A1
	Coefficient is 96228	A1
		(3)
		(3 marks)

3(a)	Attempts $\left(8-3\right)^2+\left(5--7\right)^2=...$	M1
	Writes $\left(x-3\right)^2+\left(y-5\right)^2=k$	M1
	$\left(x-3\right)^2+\left(y-5\right)^2=169$	A1
		(3)
(b)	Attempts $d^2+\left(2\sqrt{22}\right)^2=169\Rightarrow d=...$	M1
	States or uses $\left(y=\right)5+d$	dM1
	$y=14$	A1
		(3)
		(6 marks)

4(a)	 Correct shape Correct intercept Correct equation for asymptote	B1 B1 B1
		(3)
(b)	Strip width = 2.5	B1
	$\int_{-4}^{8.5}\left(3^{-\frac{1}{2}x}+4\right)\mathrm{d}x \approx \frac{5}{4}\{13+4.009+2\times(6.280+4.577+4.146+4.037)\}$ $=68.86125\left(=\text{awrt } 69\right)$	M1 A1
		(3)
(c)	(i) $\int_{-4}^{8.5}\left(3^{-\frac{1}{2}x}\right)\mathrm{d}x = 69-4\times\left(8.5--4\right)=\text{awrt } 19$	M1, A1ft
	(ii) $\int_{-4}^{8.5}\left(3^{-\frac{1}{2}x}+4\right)\mathrm{d}x + \int_{-8.5}^{4}\left(3^{\frac{1}{2}x}+4\right)\mathrm{d}x = \text{awrt } 138$	B1ft
		(3)
		(9 marks)

5(i)	$r = 0.25$	B1
	$S_\infty = \frac{1.5}{1-0.25} = 2$	M1 A1
		(3)
(ii)	(a) $\left(u_1 = 3,\right)\ u_2 = 0, u_3 = \frac{3}{2}, u_4 = 3, \ldots$	M1
	$\left(u_1 = 3\right), u_2 = 0, u_3 = \frac{3}{2}, u_4 = 3, \ldots$ and e.g. states that $u_4 = u_1$ so sequence will repeat (is periodic)	A1
		(2)
	(b) (Order) 3	A1
		(1)
	(c) $\sum_{n=1}^{70} u_n = 23 \times \left(3 + 0 + \frac{3}{2}\right) + 3 = 106.5$	M1, A1
		(2)
		(8 marks)

6 (a)	$2\log_4\left(x+3\right)=\log_4\left(x+3\right)^2$ or $\frac{1}{2}=\log_4 2$ o.e	B1
	Combines two terms e.g. $2\log_4\left(x+3\right)+\log_4 x=\log_4 x\left(x+3\right)^2$	M1
	e.g. $x\left(x+3\right)^2=2\left(4x+2\right)$	A1
	e.g. $x\left(x^2+6x+9\right)=8x+4\Rightarrow x^3+6x^2+x-4=0$ *	A1*
		(4)
(b) (i)	$x^3+6x^2+x-4=\left(x+1\right)\left(x^2+5x-4\right)$	M1
	$x=\frac{-5\pm\sqrt{25+16}}{2}\Rightarrow x=\frac{-5\pm\sqrt{41}}{2}$	dM1 A1
		(3)
(ii)	$x=\frac{-5+\sqrt{41}}{2}$	B1
		(1)
		(8 marks)

7.(a)	Attempts to use $4000 = 300 + 11d$ to find 'd'	M1
	Uses $300 + 3"d"$	M1
	Wheat production in year 4 is awrt 1310 (to the nearest 10) (tonnes)	A1
		(3)
(b)	Attempts to use $4000 = 300r^{11}$ to find 'r'	M1
	Finds $r = (1.266...)$ and MULTIPLIES this by 300	M1
	Wheat production in year 2 is awrt 380 (to the nearest 10) (tonnes)	A1
		(3)
(c)	Attempts $\frac{12}{2}\{300 + 4000\}$ or $\frac{300\left("1.266..."^{12} - 1\right)}{"1.266..." - 1}$	M1
	Finds $\frac{12}{2}\{300 + 4000\} - \frac{300\left("1.266..."^{12} - 1\right)}{"1.266..." - 1} = \left(25800 - 17935\right)$	dM1
	Difference = 7860 but allow 7870 (tonnes) (not AWRT)	A1
		(3)
		(9 marks)

8 (i)	Substitutes a value e.g. $n=6$ into n^2+3n+1 where n^2+3n+1 is **not prime**	M1
	Correct calculation for that value e.g $n^2+3n+1=55$ And conclusion "which is not prime"	A1
		(2)
(ii)	Attempts to find n^2-2 for either odds or evens E.g Attempts $(2p+1)^2-2$ or $(2p)^2-2$	M1
	Achieves either $(2p+1)^2-2=4p^2+4p-1$ or $(2p)^2-2=4p^2-2$ or gives a reason why the expression is not a multiple of 4 where required	A1
	Attempts to find n^2-2 for both odds and evens (See above)	dM1
	Achieves both $(2p+1)^2-2=4p^2+4p-1$ and $(2p)^2-2=4p^2-2$ and shows or gives reasons why these are not multiples of 4 where required With a conclusion that they are not multipl	A1*
		(4)
		(6 marks)

9 (i)	States or uses $\tan x = \dfrac{\sin x}{\cos x}$	B1
	$\sin x \tan x = 5 \Rightarrow \sin^2 x = 5\cos x \Rightarrow 1 - \cos^2 x = 5\cos x$	M1A1
	$\cos^2 x + 5\cos x - 1 = 0 \Rightarrow \left(\cos x = \right) \dfrac{-5 \pm \sqrt{29}}{2} \Rightarrow x =$ awrt $78.9°, 281.1°$	M1dM1A1
		(6)
(ii)	(a) $A = 5$	B1
		(1)
	(b) $2\theta - \dfrac{3\pi}{8} = \dfrac{3\pi}{2} \Rightarrow \theta = ...$	M1
	$\theta = \dfrac{15\pi}{16}$	A1
	y coordinate $Q = -3$ (or $2 - "A"$)	B1ft
		(3)
	(c) Sets $0 = "5" \sin\left(2\theta - \dfrac{3\pi}{8}\right) + 2 \Rightarrow \sin\left(2\theta - \dfrac{3\pi}{8}\right) = \pm\dfrac{2}{"5"}$	M1
	$\sin\left(2\theta - \dfrac{3\pi}{8}\right) = \pm\dfrac{2}{5} \Rightarrow \left(2\theta - \dfrac{3\pi}{8}\right) = \arcsin\left(\pm\dfrac{2}{5}\right) = ...$	dM1
	One of $\theta = 0.38,\ 2.4,\ 3.5,\ 5.5,\ 6.7,\ 8.6,\ 9.8....$	A1
	$\theta =$ awrt 5.51	A1
		(4)
		Total 14

10 (a)	$\left(\frac{dy}{dx}=\right) x-2187x^{-\frac{5}{2}}$	M1, A1
	Sets $x-2187x^{-\frac{5}{2}}=0 \Rightarrow x^{\frac{7}{2}}=2187 \left(\text{or e.g. } x=\left(\sqrt[7]{2187}\right)^2\right) \Rightarrow x=9$ *	dM1A1*
		(4)
(b)	e.g. $\int\left\{\frac{1}{2}x^2+1458x^{-\frac{3}{2}}-74\right\}dx=\frac{1}{6}x^3-2916x^{-\frac{1}{2}}-74x$ or $\int\left\{\frac{1}{2}x^2+1458x^{-\frac{3}{2}}-"94.5"\right\}dx=\frac{1}{6}x^3-2916x^{-\frac{1}{2}}-94.5x$	M1A1ft
	y value at P is 20.5	B1
	e.g. Area $R=\left[\frac{1}{6}x^3-2916x^{-\frac{1}{2}}-74x\right]_4^9-(9-4)\times"20.5"$ $=\left(\frac{1}{6}\times9^3-2916\times9^{-\frac{1}{2}}-74\times9\right)-\left(\frac{1}{6}\times4^3-2916\times4^{-\frac{1}{2}}-74\times4\right)-(9-4)\times"20.5"$	dM1
	$\left(-1516\frac{1}{2}+1743\frac{1}{3}-102\frac{1}{2}\right)=124\frac{1}{3}$	A1
		(5)
		Total 9